

Project Proposal

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```
library(tidyverse)
library(tidymodels)
library(readxl)
library(janitor)
library(lubridate)

# add other packages as needed

kaggle_events <- read_csv("data/global_climate_events_economic_impact_2020_2025.csv")
#probably not gonna use

raw_events <- read_excel("data/emdat_disasters_2000_2025.xlsx")

#2000 to 2025
```

Introduction

...

#2000 to 2025

Data description

...

Data processing

```

events <- raw_events |>
  clean_names()

events <- events |>
  select(disaster_type, country, start_year, start_month, start_day, end_year, end_month,
         end_day, total_deaths, no_injured, no_affected, total_affected, total_damage_000_us)

events <- events |>
  drop_na() #selecting was done before because otherwise too many rows would be dropped

events <- events |>
  mutate(
    start_date = make_date(start_year, start_month, start_day),
    end_date = make_date(end_year, end_month, end_day),
    duration_days = as.numeric(end_date - start_date)
  )

```

Analysis approach

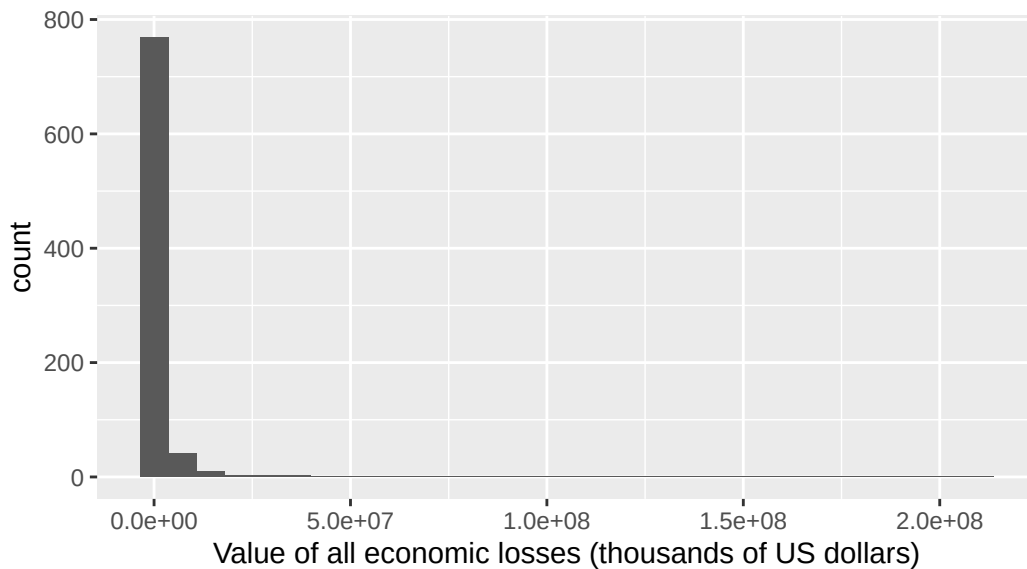
```

events |> #with outliers
  ggplot(aes(x = total_damage_000_us)) +
  geom_histogram() +
  labs(
    x = "Value of all economic losses (thousands of US dollars)",
    title = "Distribution of Economic Damage of Disasters from 2000 - 2025",
    subtitle = "With outliers"
  )

```

Distribution of Economic Damage of Disasters from 2000 – 2025

With outliers



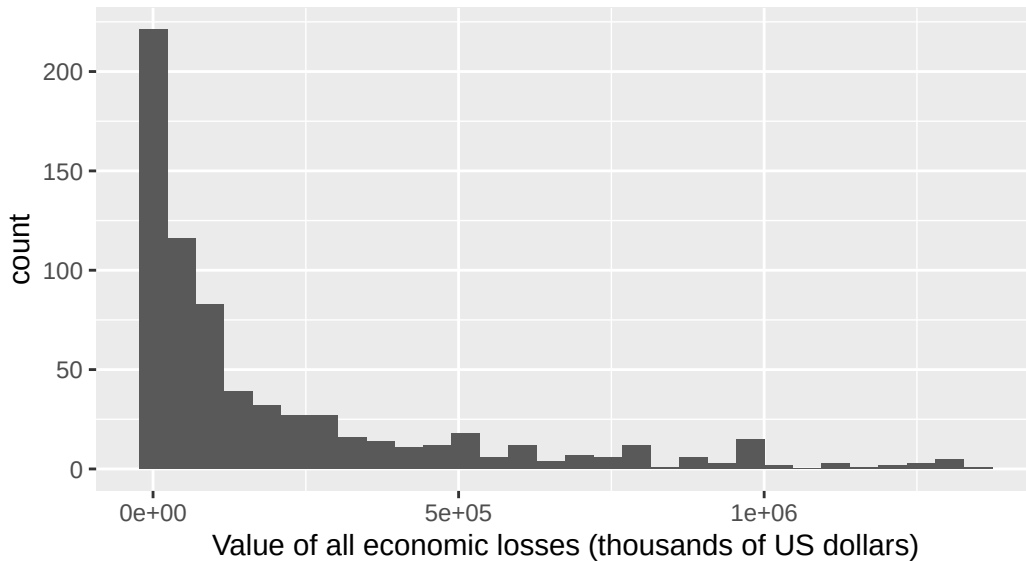
```
summary(events$total_damage_000_us)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
12	20000	109576	1519107	559750	210000000

```
events_removedOutliers <- events |>
  filter(as.integer(total_damage_000_us) < (559750 + 1.5* (559750 - 20000)) &
         as.integer(total_damage_000_us) > (20000 - 1.5 * (559750 - 20000)))

events_removedOutliers |> #without outliers
  ggplot(aes(x = total_damage_000_us)) +
  geom_histogram() +
  labs(
    x = "Value of all economic losses (thousands of US dollars)",
    title = "Distribution of Economic Damage of Disasters from 2000 - 2025",
    subtitle = "Without outliers"
  )
```

Distribution of Economic Damage of Disasters from 2000 – 2020 Without outliers



The distribution is right-skewed and unimodal. The center is best represented by the median, which is \$109,576,000, and the spread is best represented with the IQR, which is \$539,750,000 (559,750,000 - 20,000,000). There are 123 (828-705) outliers present in the upper end of our distribution.

Predictors of interests (Data from EM-DAT):

1. disaster_type - the type of disaster
2. country - country where the disaster occurred and had an impact. If multiple countries are affected, each will have an entry.
3. duration - duration in days of the disaster.
4. total_deaths - Total fatalities (deceased and missing combined)
5. no_injured - Number of people with physical injuries, trauma, or illness requiring immediate medical assistance due to the disaster
6. no_affected - Number of people requiring immediate assistance due to the disaster

We will explore multiple linear regression to predict the value (quantitative variable) of all economic losses of a disaster (in thousands of USD) with the above predictors.

Data dictionary

The data dictionary can be found [here](#) [Update the link and remove this note!]

AI Disclosure